Ex. No: 11

Date:

DESIGN OF COUNTER USING PAL

Aim:

To study and design of a counter using PAL(Programmable Array Logic).

Apparatus Required:

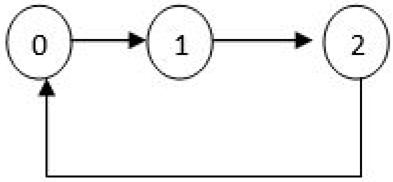
1.JK Flipflop
3. PAL
4. IC trainer kit
5. Patch chords

THEORY:

Programmable Array Logic (PAL) is a commonly used programmable logic device (PLD). It has programmable AND array and fixed OR array. Because only the AND array is programmable, it is easier to use but not flexible as compared to Programmable Logic Array (PLA). PAL's only limitation is number of AND gates. PAL consist of small programmable read only memory (PROM) and additional output logic used to implement a particular desired logic function with limited components.

Comparison with other Programmable Logic Devices: Main difference between PLA, PAL and ROM is their basic structure. In PLA, programmable AND gate is followed by programmable OR gate. In PAL, programmable AND gate is followed by fixed OR gate. In ROM, fixed AND gate array is followed by programmable OR gate array. Describing the PAL structure (programmable AND gate followed by fixed OR gate).

Mod-3 Counter Design:
State Diagram:

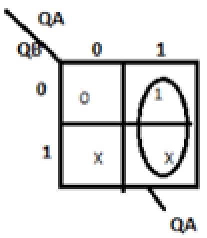


Excitation Table:

Present states		Next states		Flip-flop inputes			
Q_B	Q_A	Q_{B+1}	Q_{A+1}	J_B	K_B	J_A	K_A
0	0	0	1	0	X	1	X
0	1	1	0	1	X	X	1
1	0	0	0	X	1	0	X
1	1	0	0	X	1	X	1

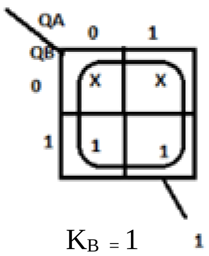
K map Implementation

For J_B



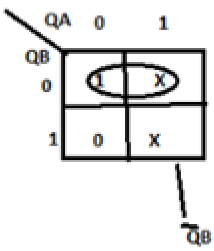
$J_B = Q_A$

For K_B



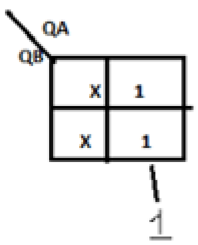
$K_B = 1$

For J_A



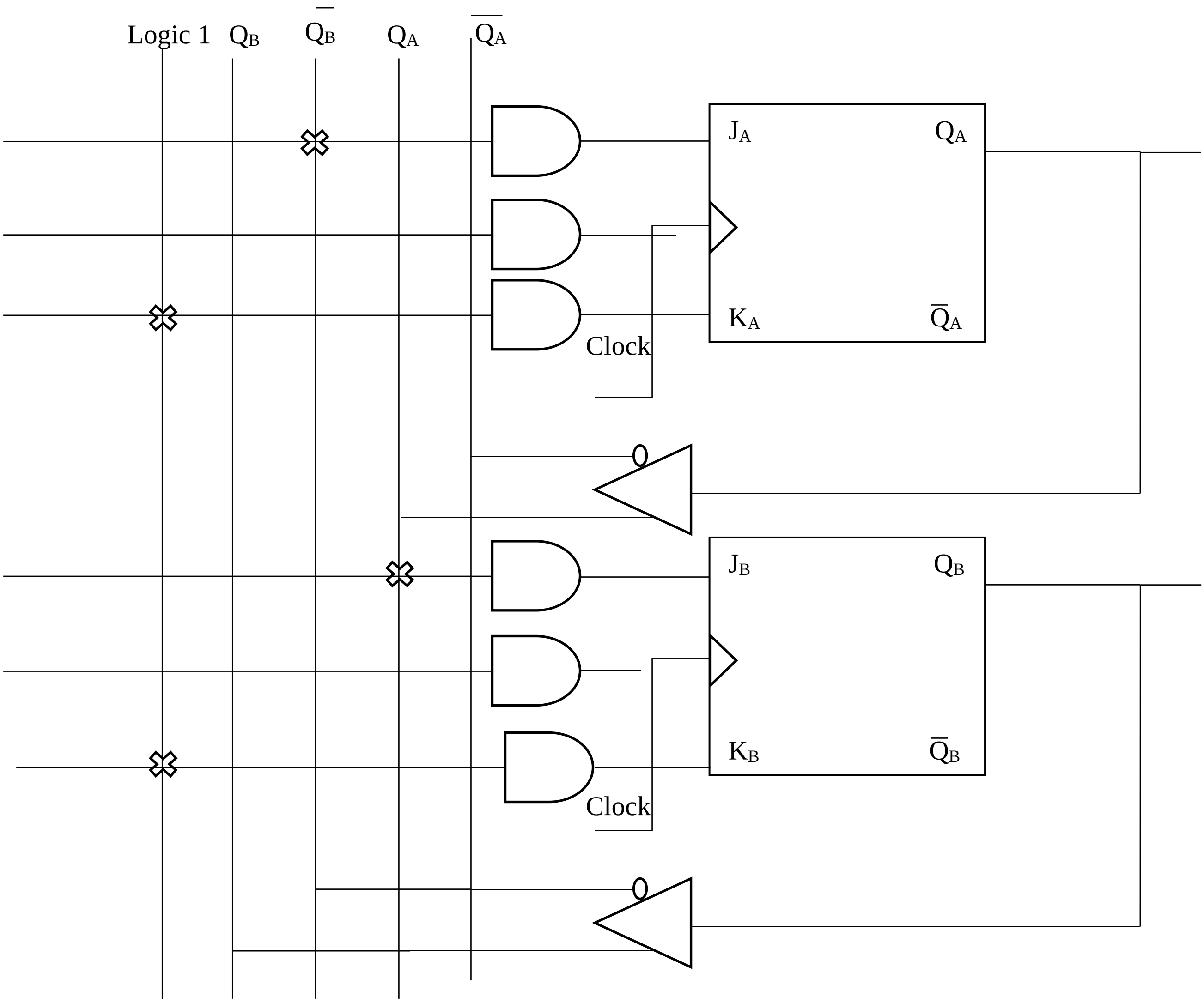
$J_A = \overline{Q}_B$

For K_A



$K_A = 1$

Circuit Diagram:



Procedure:

(i) Connections are given as per circuitdiagram.
(ii) Logical inputs are given as per circuitdiagram.
(iii) Observe the output and verify the truthtable.

Result:

Thus, Mod 3 counter using PAL was studied and designed.